

# 16.a

16. Sea  $V$  un  $K$ -espacio vectorial de dimensión finita y  $T \in L(V)$ . Hallar la representación matricial más simple de  $T$  y la base  $\mathcal{B}$  que permite obtener dicha representación matricial.

a)  $T \in L(\mathbb{R}^2) / T(x, y) = (3x - 2y, 8x - 5y)$

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• Matriz asociada

$$\begin{aligned} T(1, 0) &= (3, 8) \\ T(0, 1) &= (-2, -5) \end{aligned} \Rightarrow [T]_{\mathcal{B}} = \begin{bmatrix} 3 & -2 \\ 8 & -5 \end{bmatrix}$$

• Polinomio característico

$$\begin{aligned} P(\lambda) &= \det \begin{vmatrix} 3-\lambda & -2 \\ 8 & -5-\lambda \end{vmatrix} = (3-\lambda)(-5-\lambda) - (-16) \\ &= (\lambda+1)^2 = 0 \end{aligned}$$

$$\lambda_1 = -1; \text{ MA}(\lambda_1) = 2$$

El valor propio es único y  $\text{MA}(\lambda_1) = 2$

$$\Rightarrow T = \lambda I + N$$

$$N = T - \lambda I \Rightarrow T - (-1)I \Rightarrow T + I$$

$$N = \begin{bmatrix} 3 & -2 \\ 8 & -5 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ 8 & -4 \end{bmatrix}$$

$$\text{Im}(N) = \{(4, 8)\}$$

$$\Rightarrow \dim(\text{Im}(N)) = 1 = p$$

$$\text{Como } (k=2) \Rightarrow 2p + q = n$$

Indice de nilpotencia

$$2(1) + q = 3$$

$$q = 0 \quad (N^2 = 0)$$

$$N = \{(4, 8), (1, 0)\}$$

$$\mathcal{B} = \{(4, 8), (1, 0)\}$$

$$N(4, 8) = (0, 0) = 0(N(\mu_1)) + 0(1, 0)$$

$$N(1, 0) = (4, 8) = 1(N(\mu_1)) + 0(1, 0)$$

$$\Rightarrow [N]_{\mathcal{B}} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$T = \lambda I + N$$

$$[T]_{\mathcal{B}} = \lambda I + [N]_{\mathcal{B}}$$

$$\boxed{\lambda = -1}$$

$$[T]_{\mathcal{B}} = -1 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$[T]_{\mathcal{B}} = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}$$

$$\mathcal{B} = \{(4, 8), (1, 0)\}$$